# Marine A. Denolle

Associate Professor, Earth and Space Sciences, University of Washington | College of Environment CV — reverse chronological order; active/completed 2023–present appear first

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Google Scholar · scoped6259 · uwgeophysics6888 · denolle-lab.github.io

## Education

2008 – 2014 **Stanford University** Geophysics
- Supervisor: Dr. Gregory Beroza
- Co-supervisors: Dr. Eric Dunham, Dr. Jesse Lawrence

2007 – 2008 **Ecole Normale Supérieure - IPGP** Geophysics
- Supervisors: Dr. Satish Singh (IPGP), Dr. David Bercovici (Yale)

2006 **Ecole Normale Supérieure** Earth Sciences

2004 – 2005 **Lycée Chateaubriand, Rennes** Physics-Mathematics

## Research Impact Summary

**Research Citations:** 3,398 total (h-index: 24, i10: 45) | 2,291 last-5y ($h_5$: 22, $i10_5$: 43) | Google Scholar

**Publications:** 66 peer-reviewed | 3 Nature/Science-family (T1) | 40 AGU-flagship (T2) | 17 domain journals (T3) | 3 reviews/perspectives | 3 preprints

**Research Funding:** Lead PI: $3.1M (14 grants) | Co-PI/Co-I: $4.9M (8 grants) | Fellowships: $878K (2)

**Open-Source Software:** 372 GitHub stars | 241 forks | 18 active repos | NoisePy: 207 stars, 83 forks, 19 contributors | 1,692+ PyPI downloads/yr

**Mentoring:** 8 PhD (4 current, 4 graduated) | 11 postdocs | 15+ undergrads | 10 co-supervised grad students | 5 became faculty

**Broader Impact:** 6 workshops, 215+ seismologists trained | mlgeo-book: 11 GitHub stars

## Employment

2025 – present **Co-Founder**, Applied Environmental Intelligence – Seattle

2024 – present **Associate Professor**, University of Washington – Seattle, WA
Department of Earth and Space Sciences

2021 – 2024 **Assistant Professor**, University of Washington – Seattle, WA
Department of Earth and Space Sciences

2016 – 2021 **Assistant Professor**, Harvard University – Cambridge, MA
Department of Earth and Planetary Sciences

2014 – 2016 **Green Postdoctoral Fellow**, UC San Diego, Scripps Institution of Oceanography – La Jolla, CA
Institute of Geophysics and Planetary Physics
- Supervisor Dr. Peter Shearer

## Awards

2023 **Invited Professorship** Ecole Normale Supérieure rue d'Ulm-Paris

2023 **Data Science Fellow** eScience Institute, University of Washington

2019 **Charles F. Richter Early Career Award** Seismological Society of America

2019 **Kavli Frontiers of Science Fellow** National Academy of Sciences

2019 **Radcliffe Assistant Professorship** Institute for Advanced Study

2018 **CAREER Award** NSF

2017 **David and Lucile Packard Foundation Fellowship** David and Lucile Packard Foundation

2016 **Outstanding Reviewer** Geophysical Research Letters

2015 **Outstanding Reviewer** Geophysical Journal International

2012 **AGU Outstanding Student Paper Award** American Geophysical Union

2012 **SSA Student Presentation Award** Seismological Society of America

2010 **AGU Outstanding Student Paper Award** American Geophysical Union

## Teaching

2023 - 2025 **Computational/Advanced Seismology**, ESS 563 – U of Washington
- Taught Spring 2023, 2024, 2025
- Covers ambient noise seismology, earthquake source characterization
- Julia and Python programming for high-performance computing

2023 - 2026 **Introduction to Seismology**, ESS 412/512 – U of Washington
- Taught Winter 2023, 2025, 2026
- Covers earthquake physics, seismic waves, and hazard assessment

2022 - 2026 **Geophysics**, ESS 314 – U of Washington
- Taught 2022, 2023, 2024, 2026

2021 - 2024 **Machine Learning in the Geosciences**, ESS 469/569 – U of Washington
- Taught Fall 2021, 2022, 2023, 2024
- Open-access Jupyter Book with asynchronous teaching materials
- Course adopted by University of Arizona and UC Berkeley
- Curated geoscience-specific ML datasets for homework

2019 **Machine Learning in Earth and Planetary Sciences**, EPS 268 – Harvard
- Taught Fall 2019

2018 **Induced Seismicity**, EPS 268 – Harvard
- Taught Fall 2018

2018 - 2020 **Earthquakes and Faulting**, EPS 203 – Harvard
- Taught Fall 2018, 2020

2017 - 2020 **Earthquakes and Tectonics**, EPS 55 – Harvard
- Taught Fall 2017, 2020

2016 **Earthquake Sources**, EPS 204 – Harvard

- Taught Spring 2016

## Phd Advisees

2025 - present **Michael Hemmett**, PhD Pre-Candidate – UW ESS

Offshore geophysics

2022 - present **Manuela Koepfli**, PhD Candidate – USS

Geohazard assessment and early warning systems

- 1 publication, 1 in review, 1 in prep

2022 - present **Akash Kharita**, PhD Pre-Candidate – UW ESS

Geohazard characterization using machine learning

- 2 publications

2021 - present **Yiyu Ni**, PhD Candidate – UW ESS

Machine learning and big data seismology

- 8 publications

2019 - 2024 **Congcong Yuan**, PhD – Harvard

Time-dependent seismology, solid-fluid interaction. Next position: Cornell postdoc ⟶ Faculty position at NTU Singapore

- 4 publications

2018 - 2023 **Stephanie Olinger**, PhD – Harvard

Cryo-seismology (50% co-advised with Brad Lipovsky). Next position: Stanford Thompson Postdoctoral Fellowship ⟶ CEO of AEI

- 4 publications

2016 - 2021 **Tim Clements**, PhD – Harvard

Hydro-seismology and big-data seismology. Next position: USGS Mendenhall Postdoc ⟶ USGS Research Geophysicist

- 4 publications

2016 - 2022 **Jiuxun Yin**, PhD – Harvard

Earthquake seismology and source characterization. Next position: Caltech SCSN Postdoc Fellowship ⟶ Schlumberger

- 6 publications

## Postdocs

2024 - 2025 **Dr. Stephanie Olinger**, University of Washington

Cryoseismology, DAS, Ambient noise monitoring

- Next position: Climate Tech ⟶ CEO Applied Environmental Intelligence

2023 - 2024 **Dr. Kuan-Fu Feng**, University of Washington

Environmental Seismology, Cloud computing

- Next position: Postdoc at University of Utah

2023 - 2025 **Dr. Ethan Williams**, University of Washington

DAS

- Next position: Assistant Professor at UC Santa Cruz

2022 - 2024 **Dr. Qibin Shi**, University of Washington
DAS, Deep Learning, Earthquake Catalog building, AgroSeismology
• Now Postdoctoral Fellow at Rice University

2020 - 2022 **Dr. Laura Ermert**, Harvard University/University of Washington
Environmental seismology
• Next position: Assistant Professor at ISTerre

2019 - 2020 **Dr. Xiaotao Yang**, Harvard University
Ambient noise cross correlation
• Next position: Assistant Professor at Purdue

2019 - 2020 **Dr. Kurama Okubo**, Harvard University
Ambient noise monitoring
• Next position: Researcher at NIED, Japan

2019 - 2020 **Dr. Zhitu Ma**, Harvard University
Ambient noise cross correlation
• Next position: Assistant Professor at Tongji University, China

2018 - 2019 **Dr. Chengxin Jiang**, Harvard University
Ambient noise seismology, computational seismology
• Next position: Research Associate at Australian National University

2016 - 2017 **Dr. Chris Van Houtte**, Harvard University
Earthquake Source seismology

2016 - 2018 **Dr. Loïc Viens**, Harvard University
ambient noise seismology, ground motion prediction
• Next position Researcher at Los Alamos

## Other Graduate Student Supervision

**Note:** (*) At UW, roles are secondary advisor or primary advisor on one project/manuscript. (**) Advising resulted in a peer-reviewed publication.

**Maleen Kidiwela (*,**):** 2021-present, UW Oceanography (co-advised with William Wilcock) - DAS and offshore seismology, resulted in publication

**Zoe Krauss (*,**):** 2021-2024, UW Oceanography (co-advised with William Wilcock) - Axial Seamount hydrothermal monitoring, resulted in publication

**Parker Sprinkle (*,**):** 2021-present, UW ESS (co-advised) - Earthquake early warning and machine learning

**William Flanagan (*,**):** 2019, Harvard EPS - Volcano seismology project

**Congcong Yuan (*,**) :** 2019, USTC China - Master student visiting researcher on time-dependent seismology, resulted in publication

**Natasha Toghramadjian (*,**):** 2018-2025, Harvard EPS - Seismic attenuation and site effects, resulted in publications

**Zhuo Yang (*,**):** 2018-2021, Harvard EPS - Machine learning for seismic phase picking, resulted in publication

**Philippe Danré (*,**):** 2018, Ecole Normale Supérieure Paris - Master thesis on seismic monitoring, resulted in publication

**Thibault Pérol (*,**):** 2017, Harvard EPS - Deep learning for earthquake detection, resulted in multiple publications

**Manuel Florez (*,**):** 2017-2019, MIT - Dissertation committee member for geophysics PhD

## Graduate Student Committees

**Jones, Randall:** Atmospheric Sciences, GSR, 2025-present
**Koepfli, Manuela:** Earth and Space Sciences, Chair, 2024-present
**Ni, Yiyu:** Earth and Space Sciences, Chair, 2024-present
**Pearson, Anna Elaine Rogers:** Earth and Space Sciences, Member, 2024-present
**Kidiwela, Maleen:** School of Oceanography, Member, 2024-present
**Sangmin, Song:** School of Oceanography, GSR, 2024-present
**Sweeney, Aodhan:** Atmospheric Sciences, GSR, 2024-present
**Kharita, Akash:** Earth and Space Sciences, Advisor, 2023-present
**DeGrande, Jensen:** Earth and Space Sciences, Member, 2023-present
**Zhang, Maochuan:** School of Oceanography, GSR, 2023-present
**Chien, Mu-Ting:** Atmospheric Sciences, GSR, 2023-2024
**Ragland, John:** Electrical and Computer Engineering, GSR, 2023-2024
**Velappan, Hemalatha:** School of Environmental and Forest Sciences, GSR, 2023-present
**Velegar, Meghana S:** Applied Mathematics, GSR, 2023-2023
**Sprinkle, Parker:** Earth and Space Sciences, Member, 2022-present
**Krauss, Zoe:** School of Oceanography, GSR, 2022-2024
**Rasanen, Ryan:** Civil and Environmental Engineering, GSR, 2022-2023
**Zahn, Olivia:** Physics, GSR, 2022-2024

## International Phd Advising

**Note:** Dissertation and defense evaluating committee members
**Marius Paul Isken :** 2024 (GFZ, Germany)
**Luc Illien :** 2023 (GFZ, Germany)
**Zoe Renate :** 2023 (Universite de Lorraine, France)
**Reza Esfahani :** 2022 (GFZ, Germany)
**Kurama Okubo :** 2019 (ENS, Paris, France)

## Media Coverage

- 2025 Global-scale Database of Seismic Phases
- 2024 Ambient Field Monitoring in the Shallow Crust
- 2024 Making Big Data Look Small
- 2023 UW eScience Seminar
- 2023 Open Science in Seismology
- 2023 UW News Fiber Sensing
- 2018 Radcliffe Seminar: Where we stand in Earthquake Prediction

## Undergraduate Students

**Note:** My undergraduate advising includes research opportunities through summer programs and independent studies (ESS 499 for credit during academic year, hourly pay during summer/breaks). I mentor students toward conference presentations and publications, and support competitive fellowship applications. Legend: (*) Conference presentation | (**) Peer-reviewed publication | (***) Publication in prep | (+) NSF GRFP recipient

**Alex Rose :** 2025-present, UW Oceanography - DAS data processing research

**Anjani Mirchandi:** 2024-present, UW Applied Math - DAS data processing research

**Hiroto Bito:** 2023-2026, UW ESS - ML for earthquake catalog building and spatio-temporal forecasting

**Nicholas Wolfe:** 2023, UW ESS - Earthquake magnitude estimation project

**Informatics Capstone Team (lead Rona Guo(*)):** 2023, Nathan Limono, William Phan, Michael Yung, Matthew Herradura), UW Informatics - DAS Platform development, resulted in conference presentation

**Lucas Swanson :** 2023, UW Informatics - DAS software development

**Francesca Skene (*) :** 2022-2023, UW ESS - Surface event catalog, resulted in conference presentation

**Nick Smoczyk (*,***) :** 2022-2024, University of Minnesota - Volcano seismology using RedPy catalog, resulted in conference presentation and publication in prep

**Julian Schmitt (*,+) :** 2020-2021, Harvard EPS - Ambient noise seismology on cloud computing with Julia, BASIN project, resulted in conference presentation and NSF GRFP

**Jared Bryan (*,**,+):** 2019, Harvard EPS - Ambient noise monitoring, SCEC Internship, resulted in conference presentation, publication, and NSF GRFP

**Albert Aguilar (*):** 2018, Harvard EPS - Subduction zone seismology and data mining, SCEC, resulted in conference presentation

**Leore Lavin:** 2016, Harvard EPS - Ground motion simulations with ambient noise

**Roy Bowling (*):** 2014, Scripps IGPP - Ambient noise seismology, SCEC Internship

**Tara Larrue (*):** 2012, Stanford Geophysics - Ambient noise seismology, SURGE program

**Penprapa Wutthijuk (*):** 2011, Stanford - Ambient noise seismology, SURGE program

## Training Workshops

2025 co-PI organizer for ML and Earthquake Catalog Building workshop (35 participants)
scoped-mlscience-book.readthedocs.io

2024 Lead PI organizer for Cloud/HPC/wavefield simulations workshop (100 participants)
seisscoped.org/HPS-book/intro.html

2024 Lead organizer of cloud workshop (80 participants)
seisscoped.org/SSA-Cloud-101/intro.html

2023 Lead PI, coordinator, and instructor for HPC and Data Science
seisscoped.org/HPS-book/intro.html

2021 – present Co-organizer of ML training with Jupyter Book
geo-smart.github.io/mlgeo-book

2016 Scientific committee member and instructor

## University Service

2026 – present **Department Colloquium Committee**, University of Washington

- Help organizing the department colloquium series

2026 – present **AI@UW Advisory Committee to the VP**, University of Washington

- Advising AI strategy and initiatives across the university, commitment ~ 1hr/week

2024 – present **Co-director of UW CS4Env**, University of Washington

- Computer Science for the Environment (CS4Env) is a cross-unit initiative to promote synergies between computer science and environmental research at UW, includes organizing biweekly seminar, reviewing proposals, and organizing an annual symposium.

2024 – present **Chair of Curriculum Committee**, UW/ESS

- Reviewing proposed courses, course changes, undergraduate and graduate program.

2022 – present **Member of Curriculum Committee and Data Science Oversight Committee**, UW/ESS

- Review the proposed data science option program

2022 – 2023 **Member of Executive Committee**, UW/ESS

2022 – 2023 **Member of Research Faculty Search Committees**, UW/ESS

- Search committee for PNSN Network Manager and Igneous Processes Faculty Search

2016 – 2020 **Multiple Committee Memberships**, Harvard University

- Undergraduate Curriculum Committee, Graduate Student Council, IT Committee, Diversity Inclusion and Belonging Committee, Department Colloquium Committee

## Professional Service

2026 – present **Member of Board of Directors at Earthscope Consortium**,

2024 – 2025 **Member of Science Steering Committee at SCEC**,

2023 – 2025 **Chair of Integration and Innovation Advisory Committee at Earthscope Consortium**,

2022 – 2022 **Committee Member of Charles Richter Early Career Award Committee at Seismological Society of America**,

2021 – 2022 **Member of Data Service Standing Committee at IRIS**,

2021 – 2023 **Member of HPC Standing Committee at SCEC**,

2011 – 2012 **Chair of Graduate Student Council at Stanford University**,

## Review Panels

2020,2024 **NSF Review Panels EAR-Geophysics and GeoInformatics**,

- Reviewed 10+ proposals per panel

2016 – 2018 **USGS Review Panels Earthquake Hazards Program**,

- Reviewed 10+ proposals per panel

## Editorial Service

2025 – present **Editor at Geophysical Journal International**,

2017 – 2020 **Associate Editor at Geophysical Research Letters**,

2014 – present **Reviewer for multiple journals in geophysics and seismology**,

- >150 reviews for GJI, BSSA, Nature Communications, GRL, JGR, Science, and others

## Publications Legend

**Author Notation: Bold**: Marine A. Denolle (PI). *Italic*: graduate students, postdocs, and undergraduate students mentored by MD.

## Publications

2026 **Distributed Acoustic Sensing Records of Earthquakes and Surface Processes at Mount Rainier Volcano**
Veronica Gaete-Elgueta, *Manuela Köpfli*, Dominik Gräff, Bradley P Lipovsky, **Marine A. Denolle**, Weston A. Thelen, Brendan Pratt, Taylor R. Kenyon
(in review in Seismica)

2026 **Seismological Analysis of Contemporary and Future Alpine Fault Earthquakes Using the Southern Alps Long Skinny Array (SALSA)**
John Townend, Ilma Del Carmen Juarez-Garfias, Olivia Pita-Sllim, Calum J. Chamberlain, Emily Warren-Smith, Caroline Holden, Kasper van Wijk, **Marine Denolle**, Andrew Curtis, Hiroe Miyake
(in press in Seismological Research Letters)

2026 **Probing the Seattle Basin Edge Using a Dense Urban Nodal Array in 100 Backyards**
*Natasha Toghramadjian*, **Marine A. Denolle**, *Laura Ermert*, *Chengxin Jiang*
10.1785/0220250241 (Seismological Research Letters)

2026 **The Seismic Wavefield Common Task Framework**
Alexey Yermakov, Yue Zhao, **Marine Denolle**, *Yiyu Ni*, Philippe M. Wyder, Judah Goldfeder, Stefano Riva, *Jan Williams*, David Zoro, Amy Sara Rude, Matteo Tomasetto, Joe Germany, Joseph Bakarji, Georg Maierhofer, Miles Cranmer, J. Nathan Kutz
10.48550/arXiv.2512.19927 (arXiv)

2026 **Agroseismology: unraveling the impact of farming practices on soil hydrodynamics**
*Qibin Shi*, David Montgomery, Abigail L.S. Swann, Nicoleta C. Cristea, *Ethan Williams*, Nan You, Joe Collins, Ana Prada Barrio, Simon Jeffrey, Ana Misiewicz, Tarje Nissen-Meyer, **Marine A. Denolle**
10.48550/arXiv.2509.09821 (in review in Science - preprint on Arxiv)

2026 **A Decadal Survey of the Near-Surface Seismic Velocity Response to Hydrological Variations in Utah, United States**
*Kuan-Fu Feng*, **Marine Denolle**, Fan-Chi Lin, Tonie van Dam
(Journal of Geophysical Research: Solid Earth, 131)

2025 **Exploration of Machine Learning Methods to Seismic Event Discrimination in the Pacific Northwest**
*Akash Kharita*, **Marine Denolle**, Alexander R Hutko, J Renate Hartog, Stephen D Malone
10.48550/arXiv.2510.23795 (accepted in Seismica, pre-print on Arxiv)

2025 **A Global-scale Database of Seismic Phases from Cloud-based Picking at Petabyte Scale**
*Yiyu Ni*, **Marine Denolle**, Amanda Thomas, Alex Hamilton, Jannes Münchmeyer, Yinzhi Wang, Loïc Bachelot, Chad Trabant, David Mencin
10.26443/seismica.v4i2.1738 (Seismica, 4)

2025 **A Review of Cloud Computing and Storage in Seismology**
*Yiyu Ni*, **Marine A Denolle**, Jannes Münchmeyer, Yinzhi Wang, *Kuan-Fu Feng*, Carlos Garcia Jurado Suarez, Amanda M Thomas, Chad Trabant, Alex Hamilton, David Mencin
10.1093/gji/ggaf322 (Geophysical Journal International)

2025 **Training the Next Generation of Seismologists: Delivering Research-Grade Software Education for Cloud and HPC Computing Through Diverse Training Modalities**
**Marine A. Denolle**, Carl Tape, Ebru Bozdağ, Yinzhi Wang, Felix Waldhauser, Alice-Agnes Gabriel, Jochen Braunmiller, Bryant Chow, Liang Ding, *Kuan-Fu Feng*, Ayon Ghosh, Nathan Groebner, Aakash Gupta, *Zoe Krauss*, Amanda M. McPherson, Masaru Nagaso, Zihua Niu, *Yiyu Ni*, Rıdvan Örsvuran, Gary Pavlis, Felix Rodriguez-Cardozo, Theresa Sawi, David Schaff, Nico Schliwa, David Schneller, *Qibin Shi*, Julien Thurin, Chenxiao Wang, Kaiwen Wang, Jeremy Wing Ching Wong, Sebastian Wolf, *Congcong Yuan*
10.1785/0220240413 (Seismological Research Letters, 96)

2025 **Ambient field seismology in critical zone hydrological sciences**
**Marine A. Denolle**, *Qibin Shi*, *Tim Clements*, *Loïc Viens*, Veronica Rodriguez-Tribaldos, Fabrice Cotton
10.5802/crgeos.310 (Comptes Rendus. Géoscience, 357)

2025 **Denoising Offshore Distributed Acoustic Sensing Using Masked Auto-Encoders to Enhance Earthquake Detection**
*Q. Shi*, **M. A. Denolle**, *Y. Ni*, *E. F. Williams*, N. You
10.1029/2024JB029728 (JGR: Solid Earth, 130)

2025 **Multiplexed Distributed Acoustic Sensing Offshore Central Oregon**
*Q. Shi*, *E. F. Williams*, B. P. Lipovsky, **M. A. Denolle**, W. S. D. Wilcock, D. S. Kelley, K. Schoedl
10.1785/0220240460 (Seismological Research Letters, 96)

2024 **Wavefield reconstruction of distributed acoustic sensing: Lossy compression, wavefield separation, and edge computing**
*Y. Ni*, **M. A. Denolle**, *Q. Shi*, B. P. Lipovsky, S. Pan, J. N. Kutz
10.1029/2024JH000247 (Journal of Geophysical Research: Machine Learning and Computation, 1)

2024 **Analysing Volcanic, Tectonic, and Environmental Influences on the Seismic Velocity from 25 Years of Data at Mount St. Helens**
P. Makus, **M. A. Denolle**, C. Sens-Schönfelder, *M. Köpfli*, F. Tilmann
10.1785/0220240088 (Seismological Research Letters, 95)

2024 **Examining 22 Years of Ambient Seismic Wavefield at Mount St. Helens**
M. Köpli, **M. A. Denolle**, W. Thelen, P. Makus, S. Malone
10.1785/0220240079 (Seismological Research Letters, 95)

2024 **Impact of Temperature and Relative Humidity Variations on Coda Waves in Concrete**
F. Diewald, **M. Denolle**, J. J. Timothy, C. Gehlen
10.1038/s41598-024-69564-4 (Scientific Reports, 14)

2024 **Monitoring velocity change over 20 years at Parkfield**
*K. Okubo*, B. Delbridge, **M. Denolle**
10.1029/2023JB028084 (Journal of Geophysical Research: Solid Earth, 129)

2024 **Extended crack propagation by local nucleation and rapid transverse expansion**
T. Cochard, I. Svetlizky, G. Albertini, R. C. Viesca, S. M. Rubinstein, F. Spaepen, *C. Yuan*, **M. Denolle**, Y.-Q. Song, L. Xiao, D. A. Weitz
10.1038/s41567-023-02365-0 (Nature Physics)

2023 **Discrimination between icequakes and earthquakes in southern Alaska: an exploration of waveform features using random forest algorithm**
*A. Kharita*, **M. Denolle**, M. West
10.1093/gji/ggae106 (Geophysical Journal International)

2023 **Ocean Coupling Limits Rupture Velocity of Fastest Observed Ice Shelf Rift Propagation Event**
*S. Olinger*, B. Lipovsky, **M. Denolle**
10.1029/2023AV001023 (AGU Advances, 5)

2023 **Laboratory hydrofracture as analogs to tectonic tremors**
*C. Yuan*, T. Cochard, **M. Denolle**, J. Gomberg, A. Wech, L. Xiao, D. Weitz
10.1029/2023AV001002 (AGU Advances, 5)

2023 **Improved observations of deep earthquake ruptures using machine learning**
*Q. Shi*, **M. Denolle**
10.1029/2023JB027334 (Journal of Geophysical Research: Solid Earth, 128)

2023 **Better Together: Ensemble Learning for Earthquake Detection and Phase Picking**
*Congcong Yuan*, *Yiyu Ni*, Youzuo Lin, **Marine Denolle**
10.1109/TGRS.2023.3320148 (IEEE Transactions on Geoscience and Remote Sensing, 61)

2023 **An Object Storage for Distributed Acoustic Sensing**
*Yiyu Ni*, **Marine A. Denolle**, Rob Fatland, Naomi Alterman, Bradley P. Lipovsky, Friedrich Knuth
10.1785/0220230172 (Seismological Research Letters, 95)

2023 **Seismology in the cloud: guidance for the individual researcher**
*Z. Krauss*, *Y. Ni*, S. Henderson, **M. Denolle**
10.26443/seismica.v2i2.979 (Seismica, 2)

2023 **Curated Pacific Northwest AI-ready Seismic Dataset**
*Y. Ni*, A. Hutko, *F. Skene*, **M. Denolle**, S. Malone, P. Bodin, R. Hartog, A. Wright
10.26443/seismica.v2i1.368 (Seismica, 2)

2023 **Probing environmental and tectonic changes underneath Ciudad de México with the urban seismic field**
*L. Ermert*, E. Cabral-Cano, E. Chaussard, D. Solano-Rojas, L. Quintanar, D. Morales Padilla, E. A. Fernandez-Torres, **M. A. Denolle**
10.5194/egusphere-2022-1361 (Solid Earth (EGU))

2023 **The Seismic Signature of California's Earthquakes, Droughts, and Floods**
*T. Clements*, **M. A. Denolle**
10.1029/2022JB025553 (Journal of Geophysical Research: Solid Earth, 128)

2022 **A multitask encoder–decoder to separate earthquake and ambient noise signal in seismograms**
*J. Yin*, **M. A. Denolle**, B. He
10.1093/gji/ggac290 (Geophysical Journal International, 231)

2022 **Pronounced Seismic Anisotropy in Kanto Sedimentary Basin: A Case Study of Using Dense Arrays, Ambient Noise Seismology, and Multi-Modal Surface-Wave Imaging**
*C. Jiang*, **M. A. Denolle**
10.1029/2022JB024613 (Journal of Geophysical Research: Solid Earth, 127)

2022 **Imaging the Kanto Basin seismic basement with earthquake and noise autocorrelation functions**
*L. Viens*, *C. Jiang*, **M. A. Denolle**
10.1093/gji/ggac101 (Geophysical Journal International, 230)

2022 **Tracking the Cracking: a Holistic Analysis of Rapid Ice Shelf Fracture Using Seismology, Geodesy, and Satellite Imagery on the Pine Island Glacier Ice Shelf, West Antarctica**
*S. D. Olinger*, B. P. Lipovsky, **M. A. Denolle**, B. W. Crowell
10.1029/2021GL097604 (Geophysical Research Letters)

2022 **Detecting Elevated Pore Pressure due to Wastewater Injection Using Ambient Noise Monitoring**
*Z. Yang*, *C. Yuan*, **M. A. Denolle**
10.1785/0320210036 (The Seismic Record, 2)

2021 **The Earth's Surface Controls the Depth-Dependent Seismic Radiation of Megathrust Earthquakes**
*J. Yin*, **M. A. Denolle**
10.1029/2021AV000413 (AGU Advances, 2)

2021 **Comparing approaches to measuring seismic phase variations in the time, frequency, and wavelet domains**
*C. Yuan*, *J. Bryan*, **M. A. Denolle**
10.1093/gji/ggab140 (Geophysical Journal International, 226)

2021 **Source time function clustering reveals patterns in earthquake dynamics**
*J. Yin*, Z. Li, **M. A. Denolle**
10.1785/0220200403 (Seismological Research Letters, 92)

2021 **SeisNoise.jl: Ambient Seismic Noise Cross Correlation on the CPU and GPU in Julia**
*T. Clements*, **M. A. Denolle**
10.1785/0220200192 (Seismological Research Letters, 92)

2020 **Quiet Anthropocene, quiet Earth**
**M. A. Denolle**, T. Nissen-Meyer
10.1126/science.abd8358 (Science, 369)

2020 **SeisIO: A Fast, Efficient Geophysical Data Architecture for the Julia Language**
J. P. Jones, *K. Okubo*, *T. Clements*, **M. A. Denolle**
10.1785/0220190295 (Seismological Research Letters, 91)

2020 **NoisePy: A new high-performance python tool for ambient-noise seismology**
*C. Jiang*, **M. A. Denolle**
10.1785/0220190364 (Seismological Research Letters, 91)

2019 **Earthquakes Within Earthquakes: Patterns in Rupture Complexity**
*P. Danré*, *J. Yin*, B. Lipovsky, **M. Denolle**
10.1029/2019GL083093 (Geophysical Research Letters, 43)

2019 **Long-period ground motions from past and virtual megathrust earthquakes along the Nankai Trough, Japan**
*L. Viens*, **M. Denolle**
10.1785/0120180320 (Bulletin of the Seismological Society of America, 109)

2019 **Energetic Onset of Earthquakes**
**M. Denolle**
10.1029/2018GL080687 (Geophysical Research Letters, 46)

2019 **Relating teleseismic backprojection images to earthquake kinematics**
*J. Yin*, **M. Denolle**
10.1093/gji/ggz048 (Geophysical Journal International, 217)

2019 **Geometric Controls on Pulse-like Rupture in a Dynamic Model of the 2015 Gorkha Earthquake**
Y. Wang, **M. Denolle**, S. M. Day
10.1029/2018JB016602 (Journal of Geophysical Research, 124)

2018 **Tracking ground water using the ambient seismic field**
*T. Clements*, **M. Denolle**
10.1029/2018GL077706 ((User text suggests possible mismatch of volume/issue) Geophysical Research Letters, 123)

2018 **Complex near-surface rheology inferred from the response of greater Tokyo to strong ground motions**
*L. Viens*, **M. Denolle**, N. Hirata, S. Nakagawa
(Journal of Geophysical Research: Solid Earth, 123)

2018 **Strong Shaking Predicted in Tokyo From an Expected M7+ Itoigawa-Shizuoka Earthquake**
**M. A. Denolle**, P. Boué, N. Hirata, G. C. Beroza
(Journal of Geophysical Research: Solid Earth, 123)

2018 **Improved model fitting for the empirical Green's function approach using hierarchical models**
*C. Van Houtte*, **M. Denolle**
(Journal of Geophysical Research: Solid Earth, 123)

2018 **Tracking groundwater levels using the ambient seismic field**
*T. Clements*, **M. A. Denolle**
10.1029/2018GL077706 (Geophysical Research Letters, 45)

2018 **Convolutional neural network for earthquake detection and location**
*T. Perol*, M. Gharbi, **M. Denolle**
(Science Advances, 4)

2018 **Spatial and Temporal Evolution of Earthquake Dynamics: Case Study of the Mw 8.3 Illapel Earthquake, Chile**
*J. Yin*, **M. A. Denolle**, H. Yao
10.1002/2017JB014265 (Journal of Geophysical Research: Solid Earth, 123)

2017 **Multicomponent C3 Green's Functions for Improved Long-Period Ground-Motion Prediction**
Y. Sheng, **M. A. Denolle**, G. C. Beroza
10.1785/0120170053 (Bulletin of the Seismological Society of America, 107)

2017 **Retrieving impulse response function amplitudes from the ambient seismic field**
*L. Viens*, **M. Denolle**, H. Miyake, S. Sakai, S. Nakagawa
10.1093/gji/ggx155 (Geophysical Journal International, 210)

2016 **Beyond Basin Resonance: Characterizing Wave Propagation Using a Dense Array and the Ambient Seismic Field**
P. Boue, **M. Denolle**, N. Hirata, S. Nakagawa, G. C. Beroza
10.1093/gji/ggw205 (Geophysical Journal International, 206)

2016 **New perspective on self-similarity of shallow thrust earthquakes**
**M. Denolle**, P. M. Shearer
10.1002/2016JB013105 (Journal of Geophysical Research: Solid Earth, 121)

2015 **Dynamics of the M7.8 2015 Nepal Earthquake**
**M. Denolle**, W. Fan, P. M. Shearer
10.1002/2015GL065336 (Geophysical Research Letters, 42)

2014 **Full 3D Tomography (F3DT) for Crustal Structure in Southern California Based on the Scattering-Integral (SI) and the Adjoint-Wavefield (AW) Methods**
E.-J. Lee, P. Chen, T. H. Jordan, P. B. Maechling, **M. Denolle**, G. C. Beroza
10.1002/2014JB011236 (Journal of Geophysical Research, 119)

2014 **Long-period seismic amplification in the Kanto Basin from the ambient seismic field**
**M. Denolle**, H. Miyake, S. Nakagawa, N. Hirata, G. C. Beroza
10.1002/2014GL059425 (Geophysical Research Letters, 41)

2014 **Strong Ground Motion Prediction using Virtual Earthquakes**
**M. Denolle**, E. M. Dunham, G. A. Prieto, G. C. Beroza
10.1126/science.1245678 (Science, 343)

2013 **Ground Motion Prediction of Realistic Earthquake Sources Using the Ambient Seismic Field**
**M. Denolle**, E. M. Dunham, G. A. Prieto, G. C. Beroza
10.1029/2012JB009603 (Journal of Geophysical Research, 118)

2013 **A numeric evaluation of attenuation from ambient noise correlation functions**
J. F. Lawrence, **M. Denolle**, K. J. Seats, G. Prieto
10.1002/2012JB009513 (Journal of Geophysical Research, 118)

2012 **Solving the Surface-Wave Eigenproblem with Chebyshev Spectral Collocation**
**M. Denolle**, E. M. Dunham, G. C. Beroza
10.1785/0120110183 (Bulletin of the Seismological Society of America, 102)

2011 **On amplitude carried by the ambient seismic field**
G. A. Prieto, **M. Denolle**, J. F. Lawrence, G. C. Beroza
(Comptes Rendus Geoscience (Thematic Issue: Imaging and Monitoring with Seismic Noise), 343)

2010 **Evidence of active backthrusting at the NE Margin of Mentawai Islands, SW, Sumatra**
S. Singh, N. Hananto, A. Chauhan, H. Permana, **M. Denolle**, A. Hendriyana, D. Natawidjaja
10.1111/j.1365-246X.2009.04458.x (Geophysical Journal International, 180)

## Grants

2025 **Jerry and Linda Paros** Mult-Sensor, Multi-Geohazards Monitoring at Mt Rainier and Mountaineous regions: $750,000 (lead PI

2025 **UW CRESST - FFST** Toward Predictive Understanding of Climate-Compounded Geohazards: $300,000 (lead PI

2025 **NSF CS4All** Separating the Signal from the Noise: Promoting Alaskan students inquiry with geographically relevant seismic data and machine learning techniques: $889,766 (lead PI Lore, Denolle co-PI, $62,000 to UW), DRL-2524060

2025 **NSF RISE CAIG** Collaborative Research: CAIG: Framework for Artificial Intelligence-Enhanced Modeling of Wildfire Geohazards (FAIM-WG): Applications for postfire debris flows across the Western US: $735,074 (lead PI Istanbulluoglu, Denolle co-PI), RISE-2530591

2025 **NSF OCE** Multi-span distributed fiber sensing on the Ocean Observatories Initiative Regional Cabled Array: $880,490 (lead PI Wilcock, Denolle co-PI), OCE-2526198, technology development grant.

2025 **NSF OCE** EXTension on the ENDeavour Segment (EXTEND): Illuminating the seafloor spreading cycle: $785,819.00 (lead PI Wilcock, Denolle co-PI), OCE-2439442

2024 **NSF EAR** Collaborative Research: A seismic investigation of slow slip and fault locking along the Alaska-Aleutian subduction zone: $248,835.00 (Denolle co-PI, collaborative with lead PI Golos), EAR-2346079

2023 **NSF OCE** RAPID: Multiplexed Distributed Acoustic Sensing (DAS) at the Ocean Observatory Initiative (OOI) Regional Cabled Array (RCA): $198,069.00 (lead PI Wilcock, Denolle co-PI), OCE-2415521

2023 **Ecole Normal Supérieure de Paris** Travel Fellowship: $3,500 (Denolle PI)

2023 **SCEC** CyberTraining for Seismology: Data Science and HPC: $35,229 (Denolle PI)

2021 **David and Lucile Packard Foundation** URG Program: $50,000 (PI)

2021 **Murdock Charitable Trust Fund** UW FiberLab equipment: $950,000 (co-PI, lead PI Brad Lipovsky). Built facility for Fiber Sensing.

2021 **NSF CyberTraining** GeoSMART ML workforce development: $995,817 (co-PI). Build a course MLGEO.

2021 **NSF OAC-CSSI** Collaborative Research: Frameworks: Seismic COmputational Platform for Empowering Discovery (SCOPED): $717,441.00 (Denolle PI, lead-PI Carl Tape), OAC-2103701

2021 **NSF EAR, PREEVENTS** Collaborative Research: Cross-Validation of Empirical and Physics-based ground motion predictions: $167,804.00 (Denolle PI), EAR-2125337

2020 **SCEC grant** Aftershock patterns and co-seismic off-fault damage elucidate dynamic rupture processes on the 2019 Ridgecrest earthquake sequence: $33,307 (Denolle PI) - returned

2019 **Harvard University - David Rockefeller Center for Latin American Studies** Monitoring Seismic Hazards in Mexico City using Grillo, a Low-Cost Earthquake Early Warning System, Hardware, $85,000 (Denolle PI)

2019 **Harvard Data Science Initiative** Ambient-noise seismology using Cloud Computing: $27,210 (Denolle PI)

2018 **NSF EAR, CAREER** CAREER: Dynamics of surface rupturing thrust earthquakes: $504,315 (PI)

2018 **SCEC grant** Data Collection for Virtual Earthquakes on Cajon Pass: $28,085 (Denolle PI) - field work

2017 **David and Lucile Packard Foundation** Fellowship: $875,000 (PI)

2017 **NSF PREEVENTS** Collaborative Proposal - PREEVENTS Track 2: Cascadia Scenario Earthquakes: Source, Path, and implications for Earthquake Early Warning: $324,495.00 (Denolle PI, collaborative with lead PI Yihe Huang and Amanda Thomas), EAR-1739712

2017 **SCEC grant** Static and dynamic source parameters of global strike-slip earthquakes: $25,000 (Denolle PI)

2016 **SCEC grant** Epistemic uncertainties in ground motion prediction from virtual earthquakes: $26,173 (Denolle PI)

2015 **SCEC grant** Basin Response to Virtual Earthquakes on the San Jacinto Fault and the Itoigawa-Shizuoka Fault: $20,000 (Denolle PI)

## Research Products

2025 **Global Seismic Phase Database** Dataset
Global seismic data product from cloud-scale waveform processing. DOI.

- Scale: 4.3 billion phase picks from 1.3 PB of continuous seismic data.
- Publication link: Seismica (2025) with DOI-backed archival record.
- Supports global monitoring and data-driven geophysics workflows.

2023 **EarthML4PNW / PNW AI-Ready Seismic Dataset** Dataset
Curated ML-ready Pacific Northwest dataset product. Dataset repository.

- Publication link: Seismica (2023), DOI 10.26443/seismica.v2i1.368.
- Version-controlled release and metadata stewardship through GitHub workflows.
- Designed for reproducible benchmarking and graduate-level ML coursework.

2021 – present **Machine Learning in the Geosciences (Open-Access Jupyter Book)** Textbook
Book and course repository provide graduate-level ML curriculum, asynchronous modules, and homework data pipelines for geoscience training.

- Addresses a core gap: no canonical ML textbook existed for geoscience graduate classrooms.
- Adoption and contribution interest from peer programs (including Arizona and Berkeley).
- Ongoing community development with curated datasets and reproducible teaching workflows.

2020 – present **SeisNoise.jl** Software
Open-source Julia software for high-performance ambient-noise processing and cross-correlation. GitHub.

- Extends the Julia seismology ecosystem with core ambient-noise tooling.
- Community signal: 50 stars and 17 forks (Apr 2023 snapshot).
- Used by group members for production research workflows.

2019 – present **NoisePy** Software
Open-source Python software for large-scale ambient-noise seismology. GitHub.

- Community scale: 207 stars, 83 forks, 19 contributors (Jan 2026).
- Operational usage: taught in workshops and used for large-scale cross-correlation workflows.
- Open, version-controlled development supporting reproducible and cloud-oriented processing.

## Invited Talks

2026 **NGF Science Meeting** Conference Speaker

2026 **WCCM 2026, Munich** Conference Speaker

2026 **European Geophysical Union** Conference Speaker

2026 **CERI Memphis** Department Colloquium

2025 **SIAM Geoscience Meeting** Plenary Speaker

2025 **Workshop on Earthquake Physics and Applications of AI to Tectonic Faulting, Italy** Plenary Speaker

2025 **Civil Environmental Engineering, University of Washington** Department Colloquium

2025 **Radcliffe Institute of Advanced Studies, on the road** Short Talk

2025 **Washington University, Saint Louis** Department Colloquium

2024 **University of Southern California** Department Colloquium

2024 **University of California, Davis** Department Colloquium

2024 **Northern Arizona University** Department Colloquium

2024 **Passive imaging and monitoring in wave physics (Cargese, France)** Invited Conference Talk

2024 **Statewide California Earthquake Center** Plenary Speaker

2023 **Ecole Normale Supérieure, Paris** Séminaire Departemental

2023 **eScience Institute, University of Washington** Data Science Seminar

2023 **Sandia National Lab, GNEM seminar series** Department Colloquium

2022 **American Geophysical Union (x2)** Invited Conference Talk

2022 **University of New Mexico** Department Colloquium

2021 **University of Oregon** Seismo Colloquium

2021 **U Utah, Seismo Tea** Seismo Colloquium

2021 **University of Wisconsin** Department Colloquium

2021 **Colorado School of Mines** Department Colloquium

2021 **Mexico a traves de los sismos** Invited Conference Talk

2020 **University of Washington** Department Colloquium

2020 **UC Berkeley** Department Colloquium

2019 **Yale University** Department Colloquium

2019 **University of Washington, seismolunch** Seismo Colloquium

2019 **EGU annual meeting, Vienna** Invited Conference Talk

2019 **Michigan State University** Department Colloquium

2019 **Victoria University of Wellington, SN Jepson Lecture, New Zealand** Public Lecture

2019 **GNS-Science, New Zealand** Department Colloquium

2019 **Stanford University, Department of Geophysics** Department Colloquium

2019 **Tufts University, Civil Engineering seminar** Department Colloquium

2018 **Brown University** Department Colloquium

2018 **Ecole Normale Superieure, Paris** Department Colloquium

2017 **AGU, New Orleans** Invited Conference Talk

2017 **Columbia University, Lamont Doherty Earth Observatory** Department Colloquium

2016 **Harvard Museum of Natural History** Public Lecture

2016 **University of Oregon** Department Colloquium

2016 **University of New Hampshire, Chapman Colloquium** Department Colloquium

2016 **UC Santa Cruz, IGPP seminar** Department Colloquium

2016 **Massachusetts Institute of Technology** Department Colloquium

2015 **USGS, Menlo Park, Earthquake Hazard Program** Department Colloquium

2015 **University of Victoria, BC, Canada** Department Colloquium

2015 **Penn State, Geodynamics seminar** Department Colloquium

2015 **Harvard, Earth and Planetary Sciences** Department Colloquium

2015 **UT Austin, Solid Earth seminar** Department Colloquium

2015 **UCLA, seismology/tectonics seminar** Department Colloquium

2015 **HOKUDAN International Symposium on Active Faulting, Awaji, Japan** Invited Conference Talk

2015 **Information Theory and Applications workshop, La Jolla** Invited Conference Talk

2015 **IGPP-Scripps Institution of Oceanography, UCSD, Geophysics seminar** Department Colloquium

2015 **University of Southern California, Geophysics seminar** Department Colloquium

2014 **Strong Motion Studies for Future Mega-Quakes, DPRI, Kyoto University, Japan** Invited Conference Talk

2014 **AGU-SEG Summer Research workshop, Vancouver, Canada** Invited Conference Talk

2014 **San Diego State University** Department Colloquium

2014 **UC Santa Barbara** Department Colloquium

2014 **IGPP-Scripps Institution of Oceanography, UCSD, Geophysics seminar** Department Colloquium

2013 **AGU, Meeting of the Americas, Cancun, Mexico** Invited Conference Talk

2013 **Caltech Seismo Lab** Department Colloquium

2013 **USGS, Menlo Park** Department Colloquium

2013 **Stanford ICME seminar** Department Colloquium

2013 **Earthquake Research Institute, Tokyo University, Japan** Department Colloquium

2013 **Advanced Industrial Science and Technology, Japan** Department Colloquium

2013 **Disaster Prevention Research Institute, Japan** Department Colloquium

2012 **ACOUSTICS, France** Invited Conference Talk

2012 **Berkeley Seismo Lab** Department Colloquium

2011 **Institut de Physique du Globe de Paris, Earthquake seminar** Department Colloquium